DIRECTIONAL DERIVATIVES

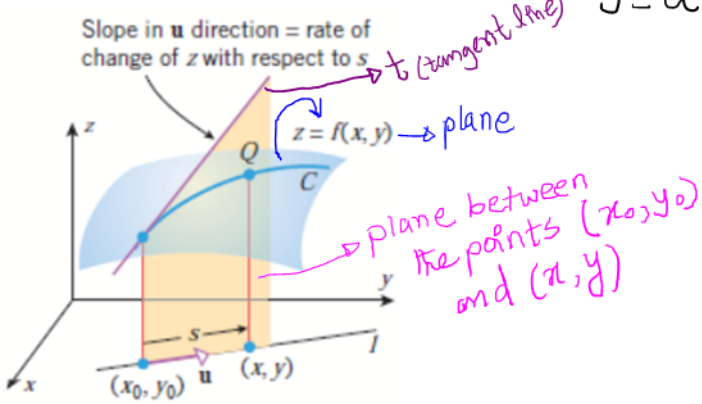
13.6.1 DEFINITION If $f(x, y)$ is a function of x and y, and if $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$ is a unit vector, then the *directional derivative of f in the direction of* $\mathbf{u}$ at (x_0, y_0) is denoted by $D_{\mathbf{u}} f(x_0, y_0)$ and is defined by

$$D_{\mathbf{u}} f(x_0, y_0) = \frac{d}{ds}\left[f(x_0 + su_1, y_0 + su_2)\right]_{s=0} \tag{2}$$

provided this derivative exists.

here u is a unit vector. magnitude = 1

$$S = d\left((x, y), (x_0, y_0)\right)$$



▲ Figure 13.6.2

The curve "c" intersects the blue & the pink plane.

The tangent line "t" to the curve "c" is the directional derivative of $z = f(x, y)$

Directional Derivative:(geometric interpretation)
rate of change of "z=f(x,y)" in the direction of "u" with respect to the distance "s"

directional derivative represents the *instantaneous rate of change of $f(x, y)$ with respect to distance in the direction of* $\mathbf{u}$ at the point (x_0, y_0).

If the point changes, the directional derivative will change as well. → consider: if (x_0, y_0) changes

The instantaneous rate of change is **the change in the rate at a particular instant**, and it is same as the change in the derivative value at a specific point. For a graph, the instantaneous rate of change at a specific point is the same as the tangent line slope. That is, it is a curve slope.

13.6.3 THEOREM

(a) *If $f(x, y)$ is differentiable at (x_0, y_0), and if $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$ is a unit vector, then the directional derivative $D_{\mathbf{u}} f(x_0, y_0)$ exists and is given by*

$$D_{\mathbf{u}} f(x_0, y_0) = f_x(x_0, y_0)u_1 + f_y(x_0, y_0)u_2 \tag{4}$$

(b) *If $f(x, y, z)$ is differentiable at (x_0, y_0, z_0), and if $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ is a unit vector, then the directional derivative $D_{\mathbf{u}} f(x_0, y_0, z_0)$ exists and is given by*

$$D_{\mathbf{u}} f(x_0, y_0, z_0) = f_x(x_0, y_0, z_0)u_1 + f_y(x_0, y_0, z_0)u_2 + f_z(x_0, y_0, z_0)u_3 \tag{5}$$

Example 1: (using Thm 13.6.3) $f_x = y$; $f_y = x$ $\quad \because f(x,y) = xy$

$f_x(1,2) = 2$; $f_y(1,2) = 1$

$D_u(1,2) = f_x(1,2)\,u_1 + f_y(1,2)\,u_2$

$= (2)\left(\frac{\sqrt{3}}{2}\right) + (1)\left(\frac{1}{2}\right) = \sqrt{3} + \frac{1}{2}$

$u = u_1 i + u_2 j = \frac{\sqrt{3}}{2}\hat{i} + \frac{1}{2}\hat{j}$

► **Example 3** Find the directional derivative of $f(x, y, z) = x^2y - yz^3 + z$ at the point $(1, -2, 0)$ in the direction of the vector $\mathbf{a} = 2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$.

Solution. The partial derivatives of f are

$$f_x(x, y, z) = 2xy, \quad f_y(x, y, z) = x^2 - z^3, \quad f_z(x, y, z) = -3yz^2 + 1$$
$$f_x(1, -2, 0) = -4, \quad f_y(1, -2, 0) = 1, \quad f_z(1, -2, 0) = 1$$

$\because a = 2i + j - 2k$

$|a| = \sqrt{2^2 + 1^2 + (-2)^2}$

$= \sqrt{9}$

Since $\mathbf{a}$ is not a unit vector, we normalize it, getting

$$\mathbf{u} = \frac{\mathbf{a}}{\|\mathbf{a}\|} = \frac{1}{\sqrt{9}}(2\mathbf{i} + \mathbf{j} - 2\mathbf{k}) = \underbrace{\frac{2}{3}}_{u_1}\mathbf{i} + \underbrace{\frac{1}{3}}_{u_2}\mathbf{j} - \underbrace{\frac{2}{3}}_{u_3}\mathbf{k}$$

Formula (5) then yields

$$D_{\mathbf{u}} f(1, -2, 0) = \underbrace{(-4)}_{f_x}\left(\frac{2}{3}\right) + \underbrace{\frac{1}{3}}_{f_y = 1} - \underbrace{\frac{2}{3}}_{f_z = 1} = -3 \quad ◄$$